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7. (a) The table shows some tests carried out to identify the ions contained in salts **A**, **B** and **C** and the observations made.

Salt	Flame test	Silver nitrate test	Name of salt
A	lilac flame	yellow precipitate	....
B	....	no change	calcium nitrate
C	....	....	barium bromide

Complete the table by giving the name of salt **A** and the observations expected for salts **B** and **C**. [3]

(b) Give the numbers of protons and electrons in a calcium ion, Ca^{2+}, and a bromide ion, Br^{-}. [2]

Ion	Number of protons	Number of electrons
Ca^{2+}	....	....
Br^{-}	....	....

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(c) Calcium nitrate solution reacts with sodium carbonate solution to form a precipitate of calcium carbonate.

$$Ca(NO_3)_2(aq) + Na_2CO_3(aq) \longrightarrow CaCO_3(s) + 2NaNO_3(aq)$$

Write an **ionic** equation for the formation of the calcium carbonate precipitate. [2]

....

(d) The solubility of barium bromide at different temperatures is shown in the table.

Temperature (°C)	Solubility (g / 100 cm^3 of water)
0	91
20	98
40	106
60	116
80	128
100	132

Calculate the mass of barium bromide that would crystallise when 500 cm^3 of a saturated solution is cooled from 80 °C to 40 °C. [2]

Mass = ... g

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